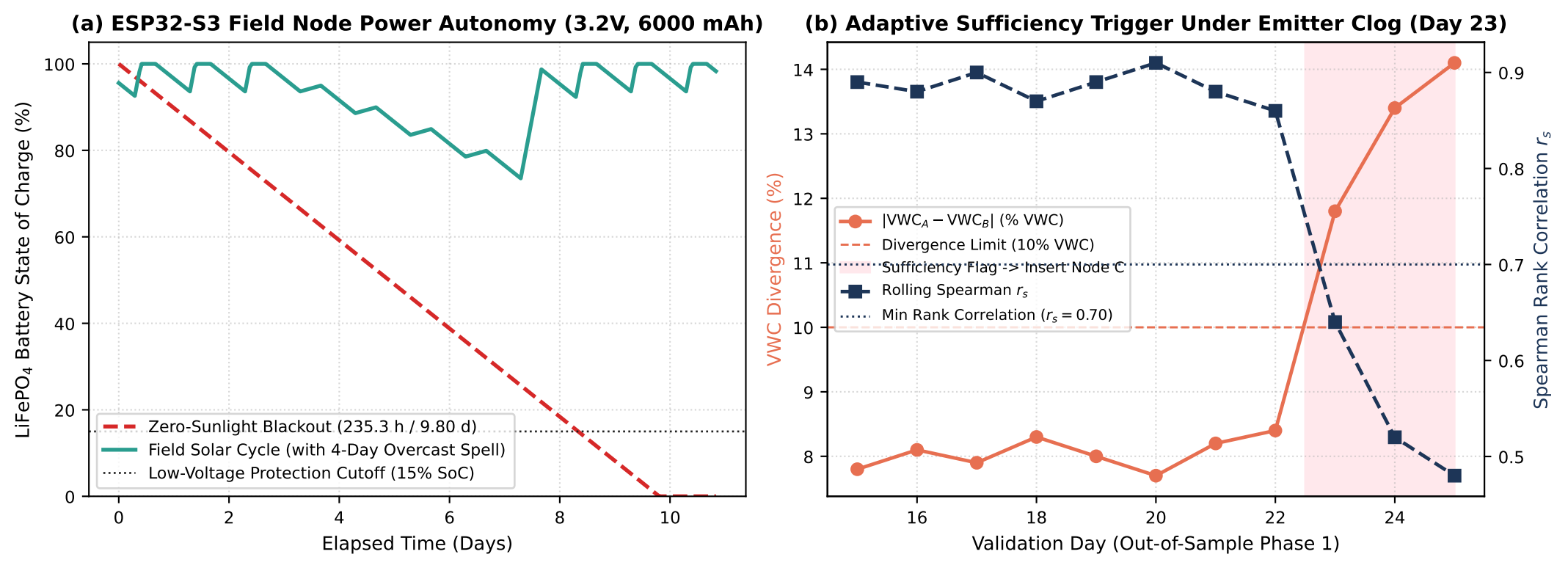
(a) ESP32-S3 Field Node Power Autonomy (3.2V, 6000 mAh)
LiFePO4 Battery State of Charge (%)
Elapsed Time (Days)
Zero-Sunlight Blackout (235.3 h / 9.80 d)
Field Solar Cycle (with 4-Day Overcast Spell)
Low-Voltage Protection Cutoff (15% SoC)
(b) Adaptive Sufficiency Trigger Under Emitter Clog (Day 23)
VWC Divergence (%)
Spearman Rank Correlation r_s
Validation Day (Out-of-Sample Phase 1)
$|VWC_A - VWC_B|$ (% VWC)
Divergence Limit (10% VWC)
Sufficiency Flag -> Insert Node C
Rolling Spearman r_s
Min Rank Correlation ($r_s = 0.70$)